

SRM VEC

SRM Nagar, Kattankulathur – 603 203

DEPARTMENT OF AIDS

QUESTION BANK

SUBJECT: AD3445 – AI1

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UNIT I – INTRODUCTION AND PROBLEM SOLVING

Introduction to AI, Intelligent Agents, Problem-solving agents, Searching for solutions, Uninformed search strategies, Breadth-first search, Depth-first search, Depth-limited search, Iterative deepening depth-first search, Bidirectional search

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define Artificial Intelligence according to the 'Acting Rationally' approach.	BTL1	Remembering
2	What is a rational agent in the context of AI?	BTL1	Remembering
3	List the four components of a PEAS description for an intelligent agent.	BTL1	Remembering
4	Distinguish between a performance measure and a utility function.	BTL2	Understanding
5	Define 'State Space' in the context of problem-solving agents.	BTL1	Remembering
6	What is the difference between an observable and a partially observable environment?	BTL2	Understanding
7	Define the term 'Branching Factor' in a search tree.	BTL1	Remembering
8	State the four criteria used to evaluate the performance of a search algorithm.	BTL1	Remembering
9	Why is Breadth-First Search (BFS) considered complete?	BTL2	Understanding
10	Mention the time and space complexity of Depth-First Search (DFS).	BTL1	Remembering
11	What is the primary motivation for using Depth-Limited Search over DFS?	BTL2	Understanding
12	Define 'Iterative Deepening Depth-First Search' (IDDFS).	BTL1	Remembering
13	How does Bidirectional Search reduce the search time complexity?	BTL2	Understanding
14	What is a 'Goal Test' in a search problem?	BTL1	Remembering
15	Differentiate between 'Tree Search' and 'Graph Search'.	BTL2	Understanding
16	What is meant by an 'Atomic Representation' of a state?	BTL1	Remembering
17	Define 'Path Cost' in the formulation of a search problem.	BTL1	Remembering
18	Explain the concept of a 'Frontier' (Open List) in search algorithms.	BTL2	Understanding
19	What is a 'Static Environment' in agent design?	BTL1	Remembering
20	Under what condition is BFS guaranteed to find the optimal solution?	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Examine the four historical approaches to Artificial Intelligence: Thinking Humanly, Thinking Rationally, Acting Humanly, and Acting Rationally. Provide examples for each.	(16)	BTL4	Analyzing
2	Categorize and explain the different types of agent programs (Simple Reflex, Model-based, Goal-based, and Utility-based) with neat architectural diagrams.	(16)	BTL4	Analyzing
3	Develop a complete PEAS description and environment characterization for the following agents: (i) Automated Taxi Driver, (ii) Satellite Image Analysis System, and (iii) Part-picking Robot.	(16)	BTL3	Applying
4	Formulate the 8-puzzle problem and the 8-queens problem as state-space search problems, specifying the states, initial state, actions, transition model, goal test, and path cost.	(16)	BTL3	Applying
5	Compare and contrast the various properties of task environments (Discrete vs. Continuous, Single vs. Multi-agent, Episodic vs. Sequential, and Known vs. Unknown) with suitable examples.	(16)	BTL4	Analyzing

6	Explain the Breadth-First Search (BFS) algorithm in detail. Illustrate its step-by-step execution on a sample graph and analyze its time and space complexity.	(16)	BTL4	Analyzing
7	Analyze the Depth-First Search (DFS) algorithm. Discuss its limitations regarding completeness and optimality, and explain how it behaves in infinite state spaces.	(16)	BTL4	Analyzing
8	Evaluate the performance of Iterative Deepening Depth-First Search (IDDFS). Prove why it combines the advantages of both BFS and DFS in terms of memory and optimality.	(16)	BTL5	Evaluating
9	Design a Bidirectional Search strategy for a route-finding problem. Discuss the challenges involved in implementing the backward search and the conditions for its success.	(16)	BTL6	Creating
10	Compare all uninformed search strategies (BFS, DFS, DLS, IDDFS, Bidirectional Search) based on the four evaluation criteria: Completeness, Time Complexity, Space Complexity, and Optimality.	(16)	BTL4	Analyzing
11	Explain the general 'Graph-Search' algorithm. Discuss how the 'explored set' helps in avoiding redundant paths and infinite loops compared to 'Tree-Search'.	(16)	BTL4	Analyzing
12	Apply the problem-solving agent framework to the 'Vacuum-Cleaner World'. Define the state space, transition model, and goal test for a 2-room environment.	(16)	BTL3	Applying
13	Critically evaluate the impact of the branching factor 'b' and depth 'd' on the memory requirements of Breadth-First Search. Why is memory a bigger bottleneck than time for BFS?	(16)	BTL5	Evaluating
14	Design a 'Learning Agent' architecture. Explain the role of each component: Learning Element, Performance Element, Critic, and Problem Generator.	(16)	BTL6	Creating
15	Analyze the Depth-Limited Search (DLS) algorithm. Explain how the choice of the depth limit 'l' affects the algorithm's completeness and optimality.	(16)	BTL4	Analyzing
16	Construct a state-space tree for the 'Missionaries and Cannibals' problem. Explain how an uninformed search strategy can be used to find a valid solution.	(16)	BTL6	Creating
17	Discuss the concept of 'Rationality' and why it is a more feasible goal for AI than 'Human-like' behavior. Justify your answer with the Turing Test perspective.	(16)	BTL5	Evaluating
18	Illustrate the difference between 'Factored' and 'Structured' representations of states with appropriate examples from the real world.	(16)	BTL3	Applying
19	Evaluate the efficiency of Uniform-Cost Search (UCS) when step costs are equal versus when they are unequal. How does it relate to BFS?	(16)	BTL5	Evaluating
20	Design an intelligent agent for a 'Smart Home Energy Management System'. Specify its PEAS, environment type, and the most suitable agent architecture.	(16)	BTL6	Creating

UNIT II – INFORMED SEARCH AND GAME PLAYING

Informed search strategies, Greedy best-first search, A search, Memory-bounded heuristic search, Local search algorithms and optimization problems, Adversarial search, Games, Optimal decisions in games, Alpha-beta pruning*

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define a heuristic function and provide an example for the 8-puzzle problem.	BTL1	Remembering
2	State the condition required for the A* search algorithm to be optimal on a graph.	BTL1	Remembering
3	Distinguish between informed search and uninformed search strategies.	BTL2	Understanding
4	What is meant by an 'admissible heuristic'?	BTL1	Remembering
5	Explain why Greedy Best-First Search is not considered optimal.	BTL2	Understanding
6	Define the concept of 'consistency' (monotonicity) in the context of heuristics.	BTL1	Remembering
7	How does the evaluation function $f(n) = g(n) + h(n)$ work in A* search?	BTL2	Understanding
8	List two memory-bounded heuristic search algorithms.	BTL1	Remembering
9	Define a local search algorithm and mention its primary advantage.	BTL1	Remembering
10	Describe the 'plateau' problem in Hill Climbing search.	BTL2	Understanding
11	What is the role of 'temperature' in the Simulated Annealing algorithm?	BTL2	Understanding
12	Define a zero-sum game in the context of adversarial search.	BTL1	Remembering
13	What is the minimax value of a node in a game tree?	BTL2	Understanding
14	State the primary objective of Alpha-Beta pruning.	BTL1	Remembering
15	Explain the meaning of 'alpha' and 'beta' parameters in pruning.	BTL2	Understanding
16	Define the 'horizon effect' in game playing.	BTL1	Remembering
17	What are the three main operators used in Genetic Algorithms?	BTL1	Remembering
18	How does Beam Search differ from standard Breadth-First Search?	BTL2	Understanding
19	Define 'Stochastic Games' and give one example.	BTL1	Remembering
20	Explain the concept of 'Optimal Decision' in a multi-player game.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the A* search algorithm in detail. Illustrate its step-by-step execution on a path-finding problem and prove its optimality given an admissible heuristic.	(16)	BTL3	Applying
2	Compare and contrast Greedy Best-First Search and A* Search. Use a specific state-space graph to demonstrate a scenario where Greedy search fails to find the shortest path while A* succeeds.	(16)	BTL4	Analyzing
3	Discuss the limitations of standard A* search regarding memory consumption. Explain how IDA* (Iterative Deepening A*) and RBFS (Recursive Best-First Search) address these limitations.	(16)	BTL4	Analyzing
4	Describe the Hill Climbing search algorithm. Analyze the common pitfalls such as local maxima, ridges, and plateaus, and suggest strategies to overcome them.	(16)	BTL4	Analyzing
5	Explain the Simulated Annealing algorithm. Evaluate how the probability of accepting a worse move changes over time and why this helps in finding a global optimum.	(16)	BTL5	Evaluating
6	Design a Genetic Algorithm to solve the 8-Queens problem. Clearly define the state representation, fitness function, selection process, crossover, and mutation operators.	(16)	BTL6	Creating

7	Explain the Minimax algorithm for two-player games. Analyze its time and space complexity and demonstrate its application on a 3-level deep game tree.	(16)	BTL4	Analyzing
8	Demonstrate the Alpha-Beta pruning process on a provided game tree. Show which branches are pruned and explain the mathematical logic behind each pruning decision.	(16)	BTL3	Applying
9	Critically evaluate the impact of move ordering on the efficiency of Alpha-Beta pruning. Discuss how perfect ordering affects the effective branching factor.	(16)	BTL5	Evaluating
10	Develop two different heuristic functions for the 15-puzzle problem. Prove whether each is admissible and consistent, and compare their performance in terms of nodes expanded.	(16)	BTL6	Creating
11	Explain the Simplified Memory-bounded A* (SMA*) algorithm. Discuss how it manages the queue when memory is full and how it avoids re-exploring discarded nodes unnecessarily.	(16)	BTL4	Analyzing
12	Apply the concept of Local Search to the Traveling Salesperson Problem (TSP). Explain how the 2-opt exchange heuristic works as a local search move.	(16)	BTL3	Applying
13	Analyze the properties of heuristics. Prove the theorem that every consistent heuristic is also admissible, and provide a counter-example for the converse.	(16)	BTL4	Analyzing
14	Explain the concept of 'Expectiminimax' for games involving chance. Describe how the evaluation function is modified to handle nodes representing random events.	(16)	BTL4	Analyzing
15	Compare and contrast different local search strategies: Hill Climbing, Simulated Annealing, and Genetic Algorithms. Discuss the suitability of each for discrete vs. continuous search spaces.	(16)	BTL4	Analyzing
16	Formulate the game of Tic-Tac-Toe as a state-space search problem. Construct a partial Minimax tree and explain how the utility values are propagated to the root.	(16)	BTL6	Creating
17	Evaluate the role of heuristic evaluation functions in real-time game playing for complex games like Chess. Discuss the trade-off between search depth and evaluation accuracy.	(16)	BTL5	Evaluating
18	Explain the Tabu Search algorithm. Discuss how the 'tabu list' prevents cycles and encourages exploration of the search space compared to basic Hill Climbing.	(16)	BTL4	Analyzing
19	Discuss the 'Quiescence Search' and 'Singular Extensions' in the context of adversarial search. Explain how they help in mitigating the horizon effect.	(16)	BTL4	Analyzing
20	Design an AI agent architecture for a game of 'Nim'. Explain the state representation, the search strategy used, and how the agent makes optimal decisions.	(16)	BTL6	Creating

UNIT III – KNOWLEDGE REPRESENTATION AND LOGIC

Logical agents, Propositional logic, Theorem proving, First-order logic, Syntax and semantics, Knowledge engineering in FOL, Inference in FOL, Forward chaining, Backward chaining, Resolution

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define a logical agent and its primary components.	BTL1	Remembering
2	Distinguish between sound inference and complete inference.	BTL2	Understanding
3	State the Modus Ponens inference rule for propositional logic.	BTL1	Remembering
4	What is meant by 'entailment' in the context of knowledge representation?	BTL1	Remembering
5	Explain the concept of a Horn Clause and its significance.	BTL2	Understanding
6	Define Skolemization and provide a simple example.	BTL1	Remembering
7	What is the role of a Knowledge Base (KB) in an AI system?	BTL2	Understanding
8	Differentiate between the syntax and semantics of First-Order Logic.	BTL2	Understanding
9	Define a 'literal' and a 'clause' in propositional logic.	BTL1	Remembering
10	What is the purpose of the Unification process in FOL?	BTL1	Remembering
11	Explain the significance of a substitution (θ) – a reference.	BTL2	Understanding
12	State the Resolution principle for theorem proving.	BTL1	Remembering
13	Define a tautology and a contradiction with examples.	BTL2	Understanding
14	What is a 'Model' in the context of logical systems?	BTL1	Remembering
15	List the basic elements of First-Order Logic (FOL).	BTL1	Remembering
16	Compare forward chaining and backward chaining briefly.	BTL2	Understanding
17	What are existential and universal quantifiers?	BTL1	Remembering
18	Explain the terms 'validity' and 'satisfiability' of a sentence.	BTL2	Understanding
19	What is Conjunctive Normal Form (CNF)?	BTL1	Remembering
20	Define a 'Ground Term' in First-Order Logic.	BTL1	Remembering

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the general architecture of a knowledge-based agent and describe how it interacts with its environment using the TELL and ASK operations.	(16)	BTL3	Applying
2	Convert the following natural language sentences into First-Order Logic (FOL) and then into Conjunctive Normal Form (CNF): 'All students who study hard pass the exam. Some students do not study hard. Therefore, some students do not pass the exam.'	(16)	BTL3	Applying
3	Perform a comparative analysis between Propositional Logic and First-Order Logic regarding their expressive power, complexity, and suitability for real-world knowledge representation.	(16)	BTL4	Analyzing
4	Describe the systematic process of Knowledge Engineering in First-Order Logic. Illustrate this process by designing a knowledge base for an electronic circuit domain.	(16)	BTL6	Creating
5	Explain the Resolution algorithm for First-Order Logic in detail. Demonstrate its application to prove a goal by contradiction given a set of axioms.	(16)	BTL3	Applying
6	Analyze the efficiency and procedural differences between Forward Chaining and Backward Chaining. Under what conditions is one preferred over the other?	(16)	BTL4	Analyzing

7	Develop the Unification algorithm for First-Order Logic and provide a step-by-step trace of unifying the predicates $P(a, x, f(g(y)))$ and $P(z, f(z), f(u))$.	(16)	BTL6	Creating
8	Critically evaluate the role of Skolem constants and Skolem functions in the process of removing existential quantifiers during CNF conversion.	(16)	BTL5	Evaluating
9	Design a First-Order Logic knowledge base for a 'Family Tree' domain. Include predicates for parent, sibling, and gender, and derive the rules for 'Grandparent' and 'Cousin'.	(16)	BTL6	Creating
10	Discuss the mathematical properties of the Resolution inference rule, specifically focusing on its soundness and refutation completeness.	(16)	BTL4	Analyzing
11	Illustrate the Wumpus World environment as a benchmark for logical agents. Represent the rules for detecting pits and the Wumpus using Propositional Logic.	(16)	BTL3	Applying
12	Evaluate the limitations of Propositional Logic in handling large-scale domains and explain how First-Order Logic overcomes these limitations through quantification.	(16)	BTL5	Evaluating
13	Demonstrate how a Backward Chaining algorithm works in a logic programming environment (like Prolog) to solve a specific query given a set of Horn clauses.	(16)	BTL3	Applying
14	Explain the transformation steps required to convert a First-Order Logic formula into Prenex Normal Form, including variable renaming and quantifier movement.	(16)	BTL4	Analyzing
15	Formulate a set of FOL sentences to represent the 'Blocks World' domain. Define predicates for 'On', 'Clear', and 'Table', and describe a goal state proof.	(16)	BTL6	Creating
16	Compare the Davis-Putnam-Logemann-Loveland (DPLL) algorithm with the Resolution method for checking the satisfiability of propositional logic sentences.	(16)	BTL4	Analyzing
17	Explain the concept of Generalized Modus Ponens (GMP). How does it facilitate efficient inference in First-Order Logic compared to standard Modus Ponens?	(16)	BTL3	Applying
18	Assess the 'Frame Problem' in knowledge representation. How do logical agents handle the challenge of representing things that do not change over time?	(16)	BTL5	Evaluating
19	Develop a logical representation for a 'University Course Registration System' using FOL. Include constraints for prerequisites and maximum enrollment.	(16)	BTL6	Creating
20	Analyze the significance of Herbrand's Theorem in the context of automated theorem proving and its relation to the semi-decidability of FOL.	(16)	BTL4	Analyzing

UNIT IV – PROBABILISTIC REASONING

Acting under uncertainty, Basic probability notation, Bayes' rule, Probabilistic reasoning, Semantics of Bayesian networks, Exact inference in Bayesian networks, Approximate inference, Hidden Markov Models

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define an atomic event in the context of probability theory.	BTL1	Remembering
2	State the product rule for two dependent random variables.	BTL1	Remembering
3	Distinguish between prior probability and posterior probability.	BTL2	Understanding
4	Write the mathematical expression for Bayes' Rule and identify its components.	BTL1	Remembering
5	What is a Conditional Probability Table (CPT) in a Bayesian Network?	BTL1	Remembering
6	Explain the concept of conditional independence between two variables.	BTL2	Understanding
7	Define the Markov property in the context of Bayesian Networks.	BTL1	Remembering
8	Differentiate between diagnostic reasoning and causal reasoning.	BTL2	Understanding
9	What is the primary objective of the Variable Elimination algorithm?	BTL2	Understanding
10	Define a Hidden Markov Model (HMM).	BTL1	Remembering
11	What is the difference between filtering and smoothing in temporal models?	BTL2	Understanding
12	State the Chain Rule for representing a joint probability distribution.	BTL1	Remembering
13	Why is exact inference in large, complex Bayesian networks computationally expensive?	BTL2	Understanding
14	Define Rejection Sampling in the context of approximate inference.	BTL1	Remembering
15	Explain the basic idea behind Likelihood Weighting.	BTL2	Understanding
16	What does the Viterbi algorithm compute in a Hidden Markov Model?	BTL1	Remembering
17	Define the transition model in a Hidden Markov Model.	BTL1	Remembering
18	What is the sensor model (observation model) in an HMM?	BTL2	Understanding
19	Define the term 'evidence' as used in probabilistic inference.	BTL1	Remembering
20	Explain why an agent might need to act under uncertainty in a real-world environment.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the fundamental components of a Bayesian Network and illustrate its semantics by constructing a network for a real-world medical diagnosis system.	(16)	BTL3	Applying
2	Compare and contrast Exact Inference (Variable Elimination) and Approximate Inference (Sampling) techniques, highlighting their trade-offs in terms of accuracy and complexity.	(16)	BTL4	Analyzing
3	Derive Bayes' Rule from the product rule of probability and demonstrate its application in a scenario where a diagnostic test result is used to predict the presence of a rare disease.	(16)	BTL3	Applying
4	Analyze the computational complexity of the Inference by Enumeration algorithm and explain how the Variable Elimination algorithm optimizes this process using factors.	(16)	BTL4	Analyzing
5	Describe the Likelihood Weighting algorithm in detail and evaluate its efficiency in handling evidence compared to the standard Rejection Sampling approach.	(16)	BTL5	Evaluating
6	Develop a complete Hidden Markov Model for a weather prediction system (states: Sunny, Rainy) based on an observer's activity (observations: Walk, Shop, Clean), defining all necessary probability matrices.	(16)	BTL6	Creating

7	Explain the Viterbi algorithm for finding the most likely sequence of hidden states in an HMM and provide a step-by-step numerical walkthrough of the algorithm.	(16)	BTL3	Applying
8	Critically discuss the concept of 'Acting under Uncertainty' and justify why probabilistic reasoning is often more robust than classical logic in partially observable environments.	(16)	BTL5	Evaluating
9	Construct a Bayesian Network for the classic 'Burglary-Alarm' problem and calculate the joint probability of a specific event using the chain rule and the network's topology.	(16)	BTL6	Creating
10	Evaluate the impact of conditional independence on the structure and storage requirements of a Bayesian Network, explaining how it reduces the number of required parameters.	(16)	BTL5	Evaluating
11	Explain the Gibbs Sampling algorithm for approximate inference and analyze why it belongs to the class of Markov Chain Monte Carlo (MCMC) methods.	(16)	BTL4	Analyzing
12	Formulate the mathematical frameworks for filtering, smoothing, and prediction in temporal models and derive the recursive estimation formula used for filtering.	(16)	BTL6	Creating
13	Analyze the performance of Rejection Sampling and Likelihood Weighting in scenarios where the evidence provided is extremely unlikely (rare events).	(16)	BTL4	Analyzing
14	Design a Bayesian Network for a car engine diagnosis system consisting of at least five nodes, and specify the Conditional Probability Tables (CPTs) for each node based on logical dependencies.	(16)	BTL6	Creating
15	Discuss the semantics of Bayesian Networks in terms of the local Markov property and the global d-separation criterion for determining independence.	(16)	BTL4	Analyzing
16	Evaluate the use of Monte Carlo methods in probabilistic reasoning, focusing on their convergence properties and their suitability for high-dimensional state spaces.	(16)	BTL5	Evaluating
17	Explain the Forward-Backward algorithm for smoothing in Hidden Markov Models and illustrate its utility in offline data analysis with a practical example.	(16)	BTL3	Applying
18	Analyze how the topology of a Bayesian Network (e.g., polytrees vs. multiply-connected networks) affects the choice and performance of inference algorithms.	(16)	BTL4	Analyzing
19	Critically examine the limitations of Hidden Markov Models in representing complex temporal dependencies and suggest how Dynamic Bayesian Networks address these issues.	(16)	BTL5	Evaluating
20	Apply the Variable Elimination algorithm to a specific multiply-connected network and demonstrate how the order of variable elimination impacts the size of intermediate factors.	(16)	BTL3	Applying

UNIT V – PLANNING AND MACHINE LEARNING

Classical planning, PDDL, State-space search for planning, Hierarchical planning, Forms of learning, Supervised learning, Learning decision trees, Evaluating and choosing the best hypothesis, Reinforcement learning

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define Classical Planning in the context of Artificial Intelligence.	BTL1	Remembering
2	List the four essential components of a PDDL action description.	BTL1	Remembering
3	Differentiate between Forward State-Space Search and Backward State-Space Search.	BTL2	Understanding
4	What is the 'Closed World Assumption' in classical planning?	BTL1	Remembering
5	Define Hierarchical Task Network (HTN) planning.	BTL1	Remembering
6	Distinguish between primitive and non-primitive tasks in hierarchical planning.	BTL2	Understanding
7	Identify the three main forms of machine learning based on the feedback available.	BTL1	Remembering
8	Explain the concept of 'Inductive Learning'.	BTL2	Understanding
9	Define 'Entropy' in the context of Decision Tree induction.	BTL1	Remembering
10	How is 'Information Gain' used to select the root node of a Decision Tree?	BTL2	Understanding
11	What is 'Overfitting' in machine learning models?	BTL1	Remembering
12	State the purpose of using a 'Validation Set' during the learning process.	BTL1	Remembering
13	Define 'Reinforcement Learning' and its core objective.	BTL1	Remembering
14	What is a 'Policy' in the framework of Reinforcement Learning?	BTL2	Understanding
15	Explain the significance of the 'Reward Function' in an RL agent.	BTL2	Understanding
16	What is 'Pruning' in Decision Trees and why is it necessary?	BTL2	Understanding
17	Define 'Hypothesis Space' in the context of supervised learning.	BTL1	Remembering
18	Contrast Classification and Regression tasks in supervised learning.	BTL2	Understanding
19	What is the 'Exploration vs. Exploitation' trade-off in Reinforcement Learning?	BTL2	Understanding
20	Define 'Ockham's Razor' as applied to hypothesis selection.	BTL1	Remembering

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the syntax and semantics of PDDL. Model the 'Sussman Anomaly' problem in the Blocks World domain using PDDL predicates and actions.	(16)	BTL3	Applying
2	Analyze the efficiency of Forward State-Space Search versus Goal-Stack Planning for complex planning problems. Provide a comparative study.	(16)	BTL4	Analyzing
3	Discuss the architecture of Hierarchical Task Networks (HTN). How does HTN planning handle large-scale real-world domains compared to classical planning?	(16)	BTL4	Analyzing
4	Illustrate the ID3 algorithm for Decision Tree induction. Use a sample dataset with at least four attributes to demonstrate the calculation of Entropy and Information Gain.	(16)	BTL3	Applying
5	Evaluate the performance of a machine learning model using K-fold Cross-Validation. Explain how this technique helps in choosing the best hypothesis.	(16)	BTL5	Evaluating
6	Formulate the Bellman Equation for Reinforcement Learning. Explain how it serves as the foundation for value-based RL algorithms like Q-Learning.	(16)	BTL6	Creating
7	Analyze the 'Bias-Variance Trade-off' in supervised learning. How does model complexity influence the generalization error?	(16)	BTL4	Analyzing

8	Design a Reinforcement Learning agent for a 'Grid World' navigation task. Define the state space, action space, and a suitable reward structure.	(16)	BTL6	Creating
9	Compare and contrast Supervised, Unsupervised, and Reinforcement Learning. Provide one industrial use-case for each and justify the choice of learning form.	(16)	BTL4	Analyzing
10	Explain the 'Planning Graph' algorithm. Discuss how it uses mutual exclusion (mutex) relations to prune the search space during plan generation.	(16)	BTL4	Analyzing
11	Critically evaluate the impact of 'Pre-pruning' and 'Post-pruning' on the accuracy and size of Decision Trees using a case study approach.	(16)	BTL5	Evaluating
12	Develop a PDDL domain and problem file for an automated warehouse robot that must move packages between different loading docks.	(16)	BTL6	Creating
13	Evaluate the use of 'Gini Index' versus 'Information Gain' as splitting criteria in Decision Trees. In which scenarios would one be preferred over the other?	(16)	BTL5	Evaluating
14	Explain the concept of 'Partial-Order Planning'. How does it differ from Total-Order Planning, and what are its advantages in terms of flexibility?	(16)	BTL4	Analyzing
15	Describe the Q-Learning algorithm in detail. Provide a step-by-step trace of how the Q-table is updated for a simple agent-environment interaction.	(16)	BTL3	Applying
16	Analyze the role of 'Inductive Bias' in learning. Why is a learner without bias unable to generalize to unseen examples?	(16)	BTL4	Analyzing
17	Design a Hierarchical Planning strategy for a 'Travel Planning' system that includes flight booking, hotel reservation, and local transport.	(16)	BTL6	Creating
18	Explain the 'Version Space' algorithm and the 'Candidate Elimination' method. How does it maintain a consistent hypothesis space?	(16)	BTL3	Applying
19	Critically evaluate the challenges of applying Reinforcement Learning in high-dimensional continuous state spaces. Suggest potential solutions.	(16)	BTL5	Evaluating
20	Compare 'Heuristic Search' in planning with 'A* Search'. How are domain-independent heuristics derived for classical planning problems?	(16)	BTL4	Analyzing

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